

Software Engineering for CS

Week 4,5

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Switch off mobile phones
during lectures,
or
put them into silent mode

Requirement???

Features, Functionality of the system?

Wanted or Needed?

Needs?

Requirement Engineering?

Gathering to Finalization.

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Introduction

Requirements form the basis for all software products

Requirements engineering is the process, which enables us to systematically determine the requirements for a software product

Requirement

Something required, something wanted or needed

- Webster's dictionary

There is a huge difference between *wanted* and *needed* and it should be kept in mind all the time

Software Requirements - 1

A complete description of *what* the software system will do without describing *how* it will do it is represented by the software requirements

Software Requirements - 2

Software requirements are complete specification of the desired external behavior of the software system to be built

They also represent External behavior of the system

Software Requirements - 3

Software requirements may be:

- Abstract statements of services and/or constraints
- Detailed mathematical functions

Software Requirements - 4

Software requirements may be:

- Part of the bid of contract
- The contract itself
- Part of the technical document, which describes a product

IEEE Definition

A condition or capability that must be met or possessed by a system...to satisfy a contract, standard, specification, or other formally imposed document

- IEEE Std 729

Sources of Requirements

Stakeholders

- People affected in some way by the system

Documents

Existing system

Domain/business area

Sources of Requirements

Stakeholders

- People affected in some way by the system
 - Owner (Investor)
 - Users
 - The team working on the software and responsible for maintaining it
 - External
 - Competitors
 - News agencies
 - Influencers
 - Government

Documents

Existing system

Domain/business area

Levels of Software Requirements

Stakeholders describe requirements at different levels of detail

- *“What versus How”*
- *“One person’s floor is another person’s ceiling”*

Importance of Software Requirements

The hardest single part of building a software system is deciding what to build...No other part of the work so cripples the resulting system if done wrong. No other part is difficult to rectify later, *Fred Brooks*



What the customer
really needed



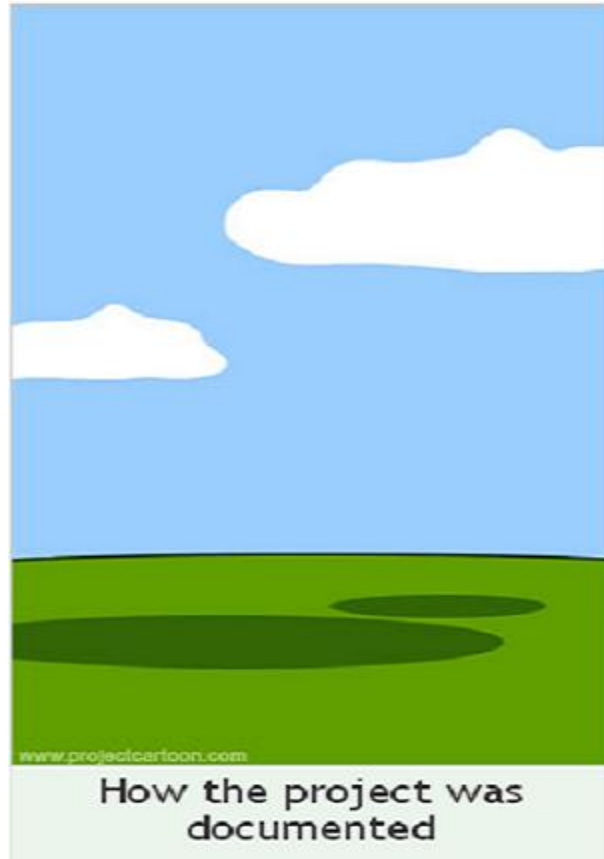
How the customer
explained it



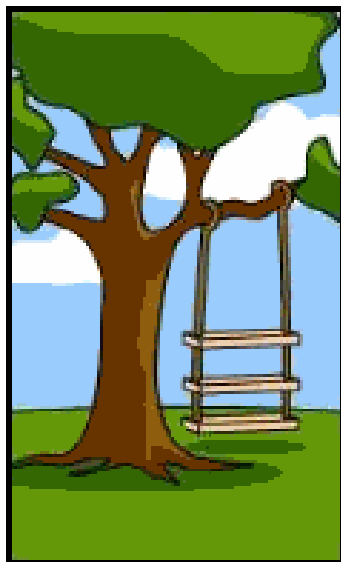
How the project leader
understood it



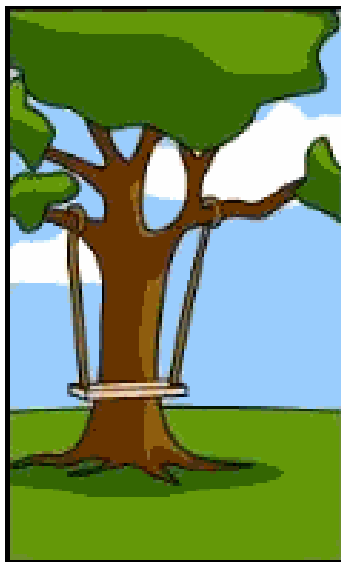
How the analyst
designed it



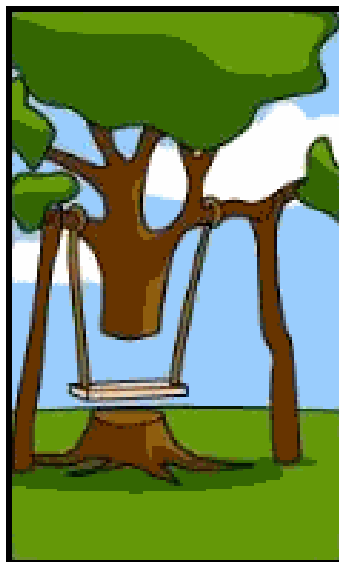
Here is the complete
story



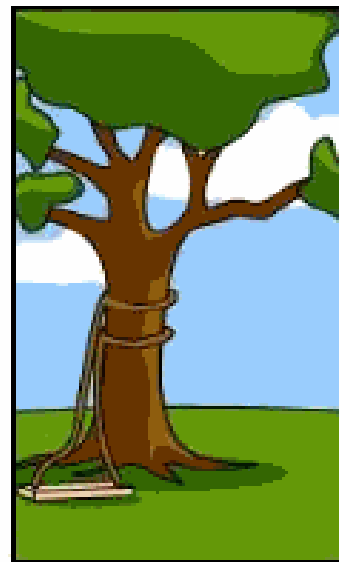
How the customer explained it



How the Project Leader understood it



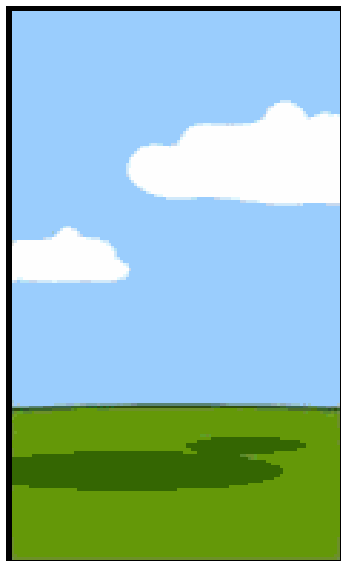
How the Analyst designed it



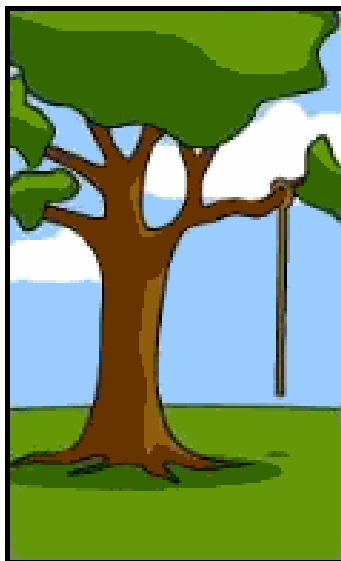
How the Programmer wrote it



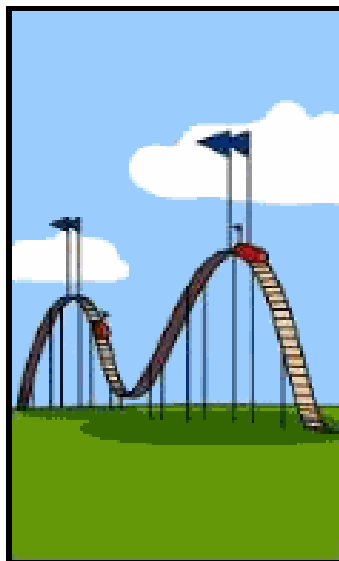
How the Business Consultant described it



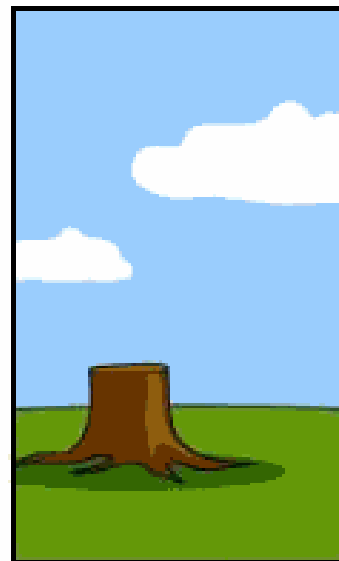
How the project was documented



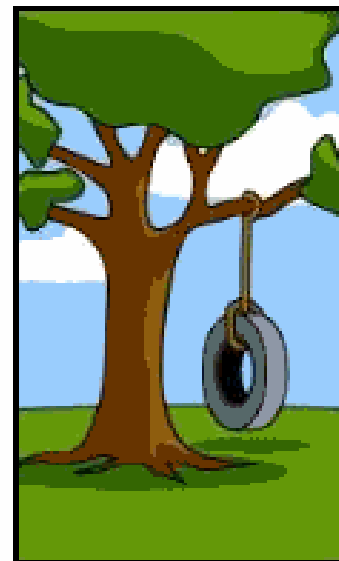
What operations installed



How the customer was billed



How it was supported



What the customer really needed

Examples of Requirements - 1

The system shall maintain records of all payments made to employees on accounts of salaries, bonuses, travel/daily allowances, medical allowances, etc.

Examples of Requirements - 2

The system shall interface with the central computer to send daily sales and inventory data from every retail store

Examples of Requirements - 3

The system shall maintain records of all library materials including books, serials, newspapers and magazines, video and audio tapes, reports, collections of transparencies, CD-ROMs, DVDs, etc.

Examples of Requirements - 4

The system shall allow users to search for an item by title, author, or by International Standard Book Number

The system's user interface shall be implemented using a web browser

Examples of Requirements - 5

The system shall support at least twenty transactions per second

The system facilities which are available to public users shall be demonstrable in ten minutes or less

Kinds of Software Requirements

Kinds of Software Requirements

Functional requirements

Non-functional requirements

Domain requirements

Inverse requirements

Design and implementation constraints

Functional Requirements - 1

Statements describing what the system does

Functionality of the system

Functional Requirements - 2

Statements of services the system should provide

- Reaction to particular inputs
- Behavior in particular situations

Functional Requirements - 3

Sequencing and parallelism are also captured by functional requirements

Abnormal behavior is also documented as functional requirements in the form of exception handling

Functional Requirements - 4

Functional requirements should be complete and consistent

Customers and developers usually focus all their attention on functional requirements

Functional Requirements Example # 1

The system shall solve a quadratic equation using the following formula

$$x = \frac{-b \pm \sqrt{b^2 - 4 * a * c}}{2 * a}$$

Functional Requirements Example # 2

The user shall be able to search either the entire database of patients or select a subset from it (admitted patients, or patients with asthma, etc.)

Functional Requirements Example # 3

The system shall provide appropriate viewers for the user to read documents in the document store

Functional Requirements Example # 4

Every order shall be allocated a unique identifier (ORDER_ID) which the user shall use to access that order

Functional Requirements Example # 5

The system shall allow customers to return non-perishable items within fifteen days of the purchase. A customer must present the original sale receipt to return an item

Comments on Examples

Notice the level of detail in different requirements described above.
Some are very detailed compared to others

Comments on Examples

Notice the ambiguity in the requirement, which uses the term 'appropriate viewers'

This requirement does not mention the formats of documents and types of viewers, which can be used

Comments on Examples

Notice the ambiguity in the requirement for solving the quadratic equation. The requirement does not speak about the possibility when the value of 'a' is zero

$$x = (-b \pm \sqrt{b^2 - 4*a*c}) / 2*a$$

Comments on Examples

Incomplete and ambiguous requirements are open to multiple interpretations and assumptions

This can lead to the development of poor quality, or faulty, software products

Non-Functional Requirements - 1

Most non-functional requirements relate to the system as a whole. They include constraints on timing, performance, reliability, security, maintainability, accuracy, the development process, standards, etc.

Non-Functional Requirements - 2

They are often more critical than individual functional requirements

Capture the emergent behavior of the system, that is they relate to system as a whole

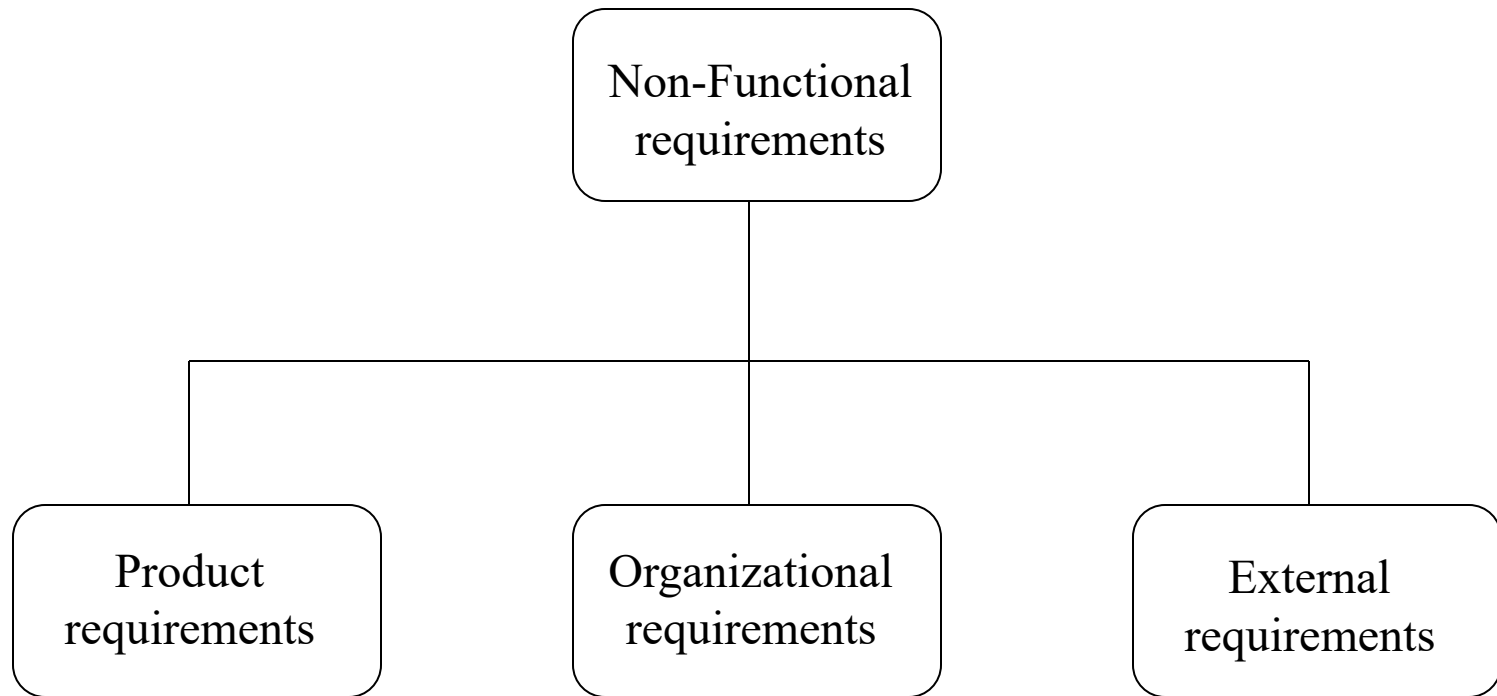
Many non-functional requirements describe the quality attributes of the software product

Non-Functional Requirements - 3

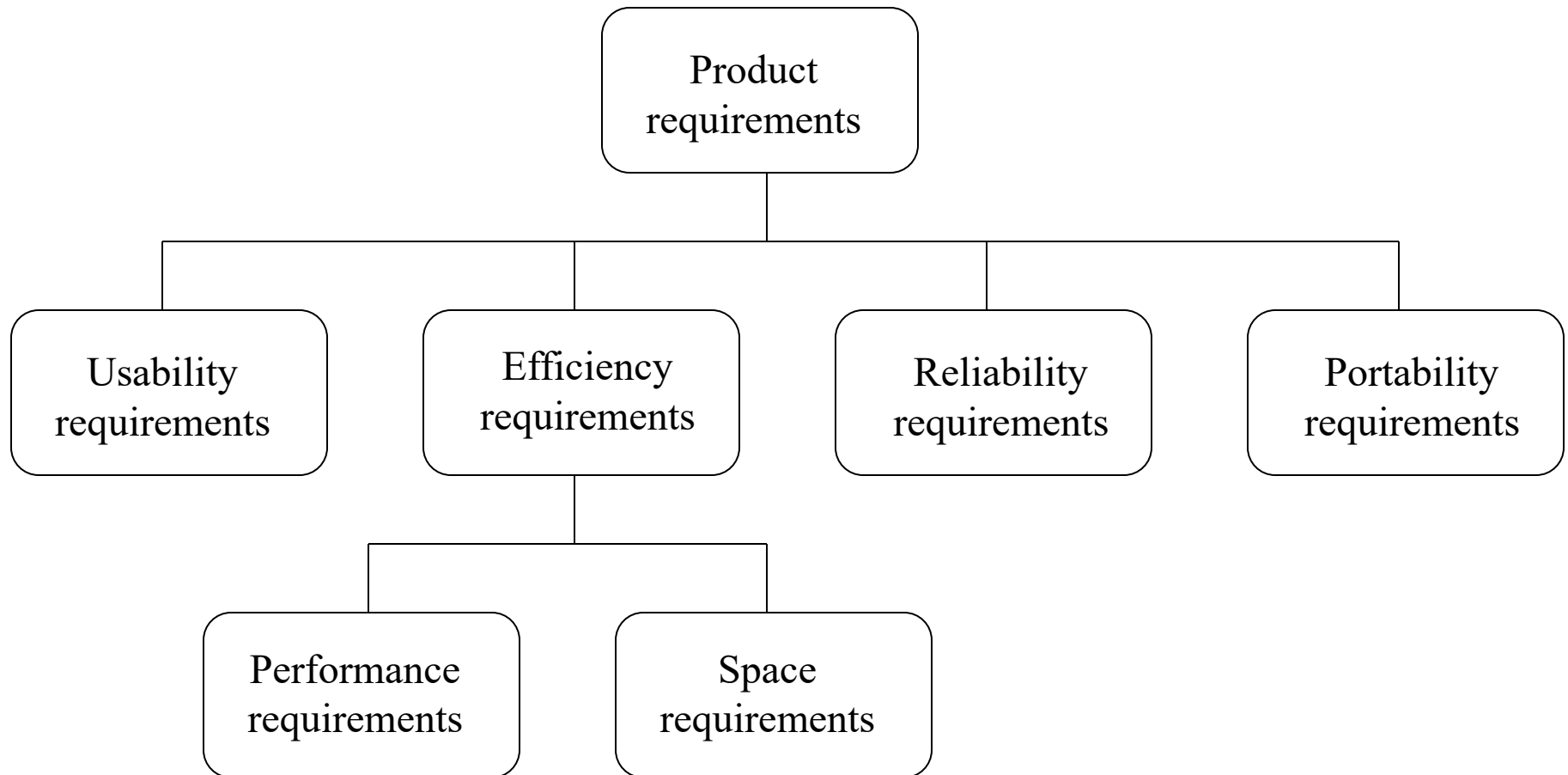
For example, if an aircraft system does not meet reliability requirements, it will not be certified as 'safe'

If a real-time control system fails to meet its performance requirements, the control functions will not operate correctly

Non-Functional Requirements - 4



Product Requirements - 1

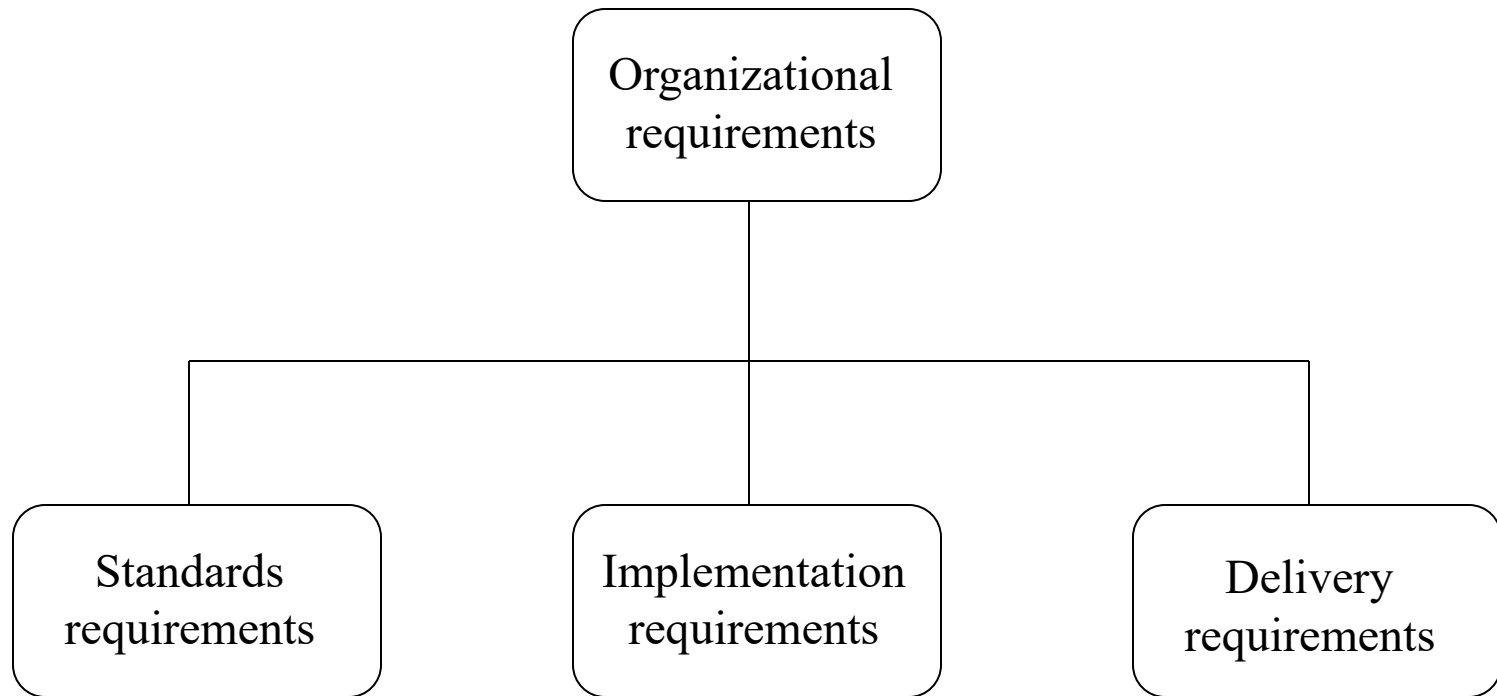


Product Requirements

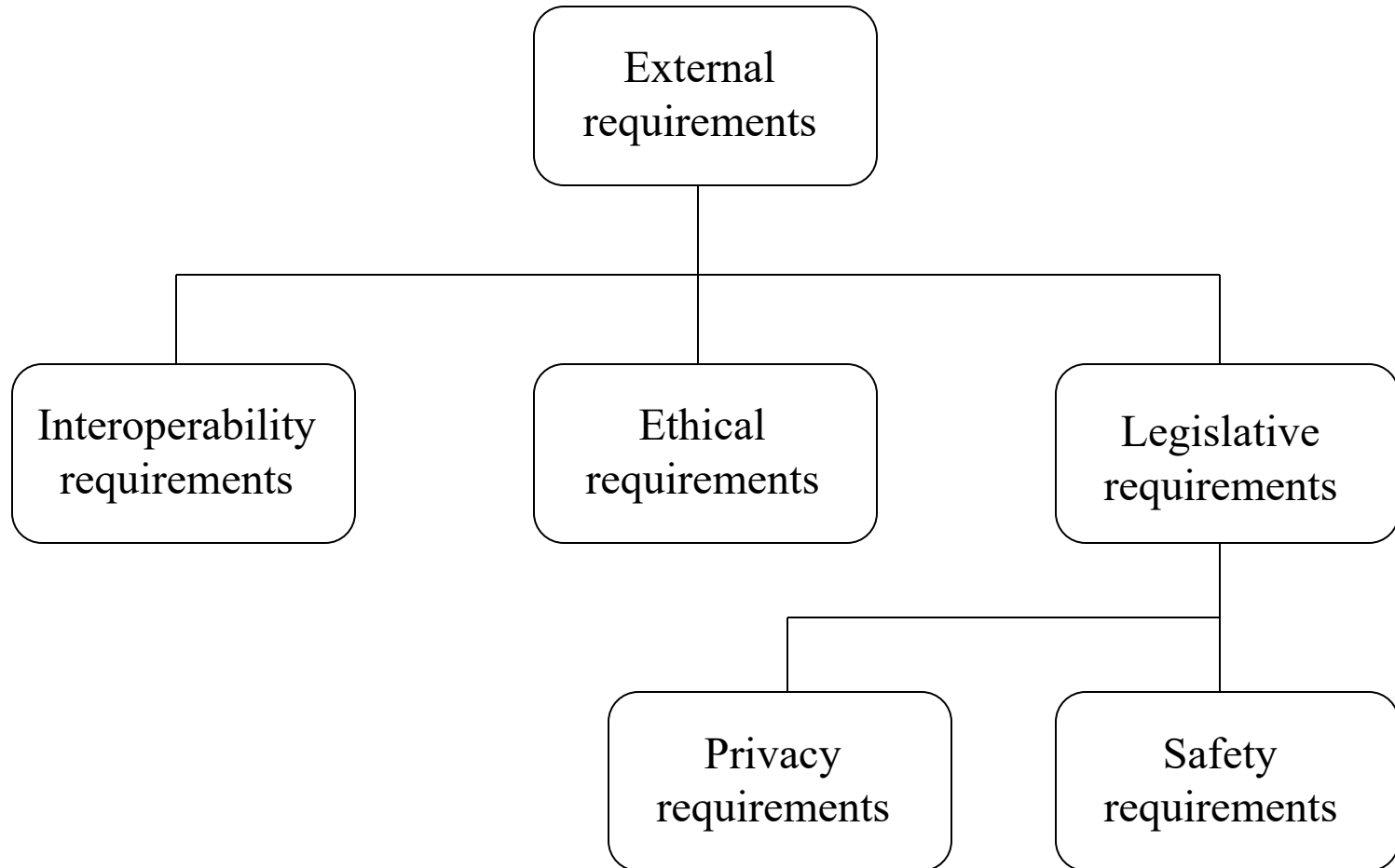
The system shall allow one hundred thousand hits per minute on the website

The system shall not have down time of more than one second for continuous execution of one thousand hours

Organizational Requirements - 1



External Requirements - 1



External Requirements - 2

The system shall not disclose any personal information about members of the library system to other members except system administrators

The system shall comply with the local and national laws regarding the use of software tools

Observations on Non-Functional Requirements - 1

Non-functional requirements are written to reflect general goals for the system. Examples include:

- **Ease of use**
- **Recovery from failure**
- **Rapid user response**

Observations on Non-Functional Requirements - 2

Goals are open to misinterpretation

Objective verification is difficult

Distinction between functional and non-functional is not always very clear

Observations on Non-Functional Requirements - 3

Non-functional requirements should be written in a quantitative manner as much as possible, which is not always easy for customers

For some goals, there are no quantitative measures, e.g., maintainability

Observations on Non-Functional Requirements - 4

Goals can be useful to designers and developers, as they give clues to them about priorities of the customers

Observations on Non-Functional Requirements - 5

Chances of conflicts within non-functional requirements are fairly high, because information is coming from different stakeholders. For example, different stakeholders can give different response times or failure tolerance levels, etc.

Observations on Non-Functional Requirements - 6

Some negotiations must be done among different stakeholders, to achieve an agreement in these situations

Observations on Non-Functional Requirements - 7

Non-functional requirements should be highlighted in the requirements document, so that they can be used to build the architecture of the software product

Domain Requirements - 1

Requirements that come from the application domain and reflect fundamental characteristics of that application domain

Can be functional or non-functional

These requirements, sometimes, are not explicitly mentioned, as domain experts find it difficult to convey them. However, their absence can cause significant dissatisfaction

Domain Requirements - 2

Domain requirements can impose strict constraints on solutions. This is particularly true for scientific and engineering domains

Domain-specific terminology can also cause confusion

Inverse Requirements

They explain what the system shall **not** do. Many people find it convenient to describe their needs in this manner

These requirements indicate the indecisive nature of customers about certain aspects of a new software product

Design and Implementation Constraints - 1

They are development guidelines within which the designer must work, which can seriously limit design and implementation options

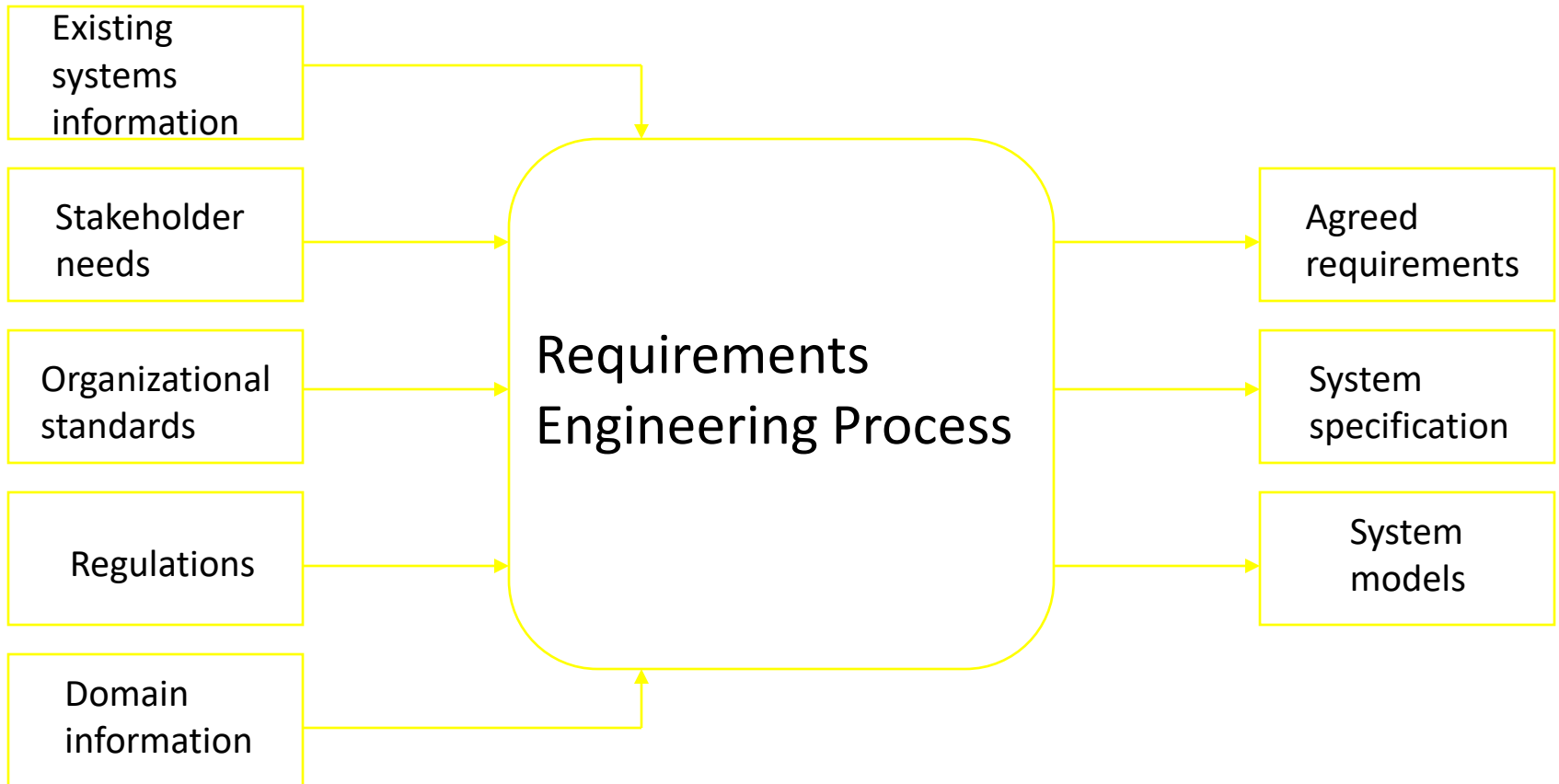
Can also have impact on human resources

Requirements Engineering Process

The process(es) involved in developing system requirements is collectively known as

Requirements Engineering Process

RE Process - Inputs and Outputs



RE Process – Inputs

It includes:

Existing system information

- Information about the functionality of systems to be replaced
- Information about other systems, which interact with the system being specified

RE Process – Inputs

Stakeholder needs

- Description of what system stakeholders need from the system to support their work

Organizational standards

- Standards used in an organization regarding system development practice, quality management, etc.

RE Process – Inputs

Regulations

- External regulations such as health and safety regulations, which apply to the system

Domain information

- General information about the application domain of the system

RE Process – Outputs

It includes

Agreed requirements

- A description of the system requirements, which is understandable by stakeholders and which has been agreed by them

RE Process – Outputs

System specification

- This is a more detailed specification of the system, which may be produced in some cases

RE Process – Outputs

System models

- A set of models such as a data-flow model, an object model, a process model, etc., which describes the system from different perspectives

Thanks